conditions. Travel booking platforms rely on APIs to aggregate data from airlines, hotels, and car rental services, providing users with comprehensive travel options. APIs provide enhanced accessibility to services and information, facilitating innovation for underserved populations and empowering developers with localised solutions.

APIs play a crucial role in enabling the modular approach to software development in the realm of microservices. Microservices architecture breaks down applications into smaller, independent services that can be developed, deployed, and scaled independently. Each microservice communicates with others through APIs, ensuring that they work together seamlessly. This approach enhances flexibility, scalability, and resilience, allowing organisations to respond quickly to changing business needs and technological advancements.

APIs are also indispensable in event-driven architecture (EDA), a design paradigm in which the program's flow is determined by events such as user actions, sensor outputs, or messages from other programs. APIs facilitate communication between event producers and consumers, enabling real-time data processing and decision-making. This architecture is instrumental in high responsiveness and scalability scenarios, such as financial trading systems, IoT applications, and real-time analytics.

The emerging trend of 'API-as-a-product', where APIs themselves are treated as valuable commercial offerings, also holds potential. If adopted by local enterprises or community organisations in underserved areas, this approach can empower them to offer their unique local data or specialised services to a broader market. For instance, a local agricultural cooperative could expose real-time local market price information via an API. This data could then be used by other developers or businesses to build valuable services for farmers, such as pricing tools or market access platforms. This not only creates potential revenue streams for the cooperative but also disseminates valuable local information, empowering the community and fostering self-sufficiency and local innovation.

The AI-API connect

Artificial intelligence (AI) is fundamentally transforming the API industry. As noted by Marco Palladino, CTO of Kong, "There is no AI without APIs," underscoring the symbiotic relationship between these technologies. AI is influencing API creation, definition, and description, with AI agents increasingly relying on API usage to perform tasks and automate processes. This includes AI guiding the development and consumption of APIs, and APIs themselves becoming more machine-readable through schema-based generation, which facilitates interaction with large language models (LLMs). Specialised AI gateways are also emerging to handle AI-specific features and security requirements.

The rise of AI agents consuming APIs to automate access to digital services presents a dual prospect. On one hand, it could simplify interactions for users with low digital literacy, making services more accessible. On the other hand, biases embedded within AI agents or the APIs they interact with could disproportionately harm underserved communities. If an AI agent used for accessing social services, for example, carries inherent biases from its training data, it could unfairly deny or misdirect vulnerable users, thereby reinforcing existing inequalities. This underscores the need for robust governance, ethical considerations, and responsible AI regulation.

Data privacy and security within integrations are also critical, demanding compliance with regulations such as GDPR and CCPA. For APIs to effectively bridge the digital divide, they must be inherently trustworthy. Weak governance frameworks or security vulnerabilities can disproportionately affect vulnerable users, who may have limited recourse or awareness of the associated risks. Such incidents can erode trust and hinder the adoption of potentially beneficial digital services, thereby undermining inclusion efforts.

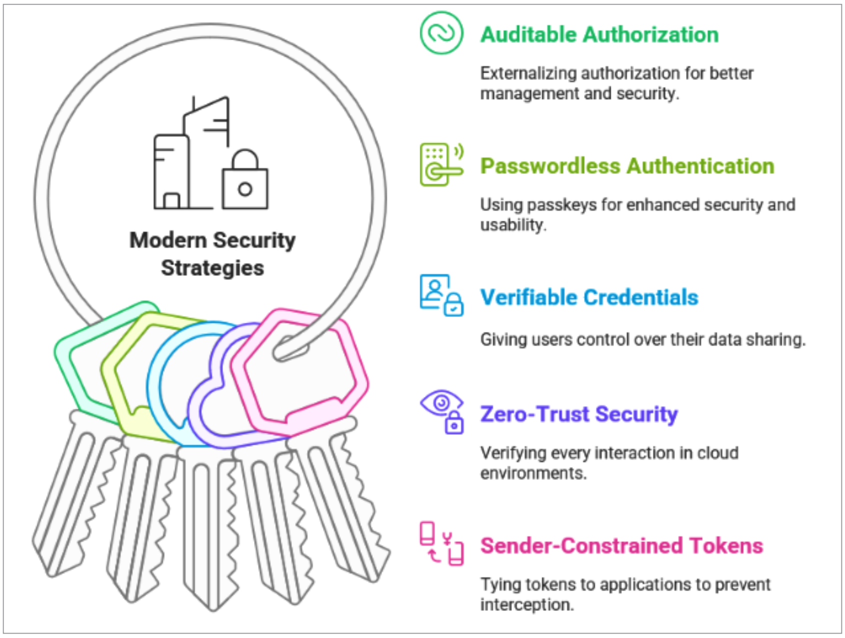

Figure 2: Modern security strategies